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A Scenario Analysis of Production, Employment, and Sales for Photovoltaic (PV) Module Manufacturing in Eskisehir, Türkiye

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Abstract

This study develops a calibrated scenario analysis of production, employment, and sales for solar photovoltaic (PV) module manufacturing in Eskisehir, Türkiye. A Cobb–Douglas production function with an energy input is specified and calibrated rather than econometrically estimated. A firsthand household electricity survey conducted in Eskisehir in 2013 is used to characterize potential local demand, not to validate the production parameters. Meanwhile, the relationship between "Leisure time" and "Work" is theoretically linked to the "Output" under free-market conditions and profit maximization. This is achieved by expressing Output in terms of "Work". Therefore, the novelty lies in defining the output as a function of "Work" and its counterpart, "Leisure time", using the cosine function in the mathematical definition of "Work" via the scalar product as $W = \vec{F} \cdot \vec{d}$, and defined as W = F d cos θ. Here, the analogues of F, d, and cos θ are Energy (E), Labor × Capital (L × K), and Human capital (H), respectively. With this adaptation to energy economics, cos θ is used as a dimensionless, calibrated index of labor effectiveness. The representation is an expository normalization and not a utility-based theory of labor supply: Equation (14) is an identity by construction rather than a testable behavioral relation. The reference plant is assumed to employ 15 production workers and two support staff. This workforce is an assumption of scale and is not derived as an economic cost optimum. At that assumed scale, the ideal case assembles 21.2 three-kilowatt module sets per day and the benchmark case 13.0 with the same workforce, the effectiveness index scaling output without changing the assumed scale. Valued at the complete installed-system price, the revenue net of labor cost is 56.8 million EUR per year in the ideal case and 34.8 million EUR in the benchmark case over a thirty-year horizon; valued at the module share of that price, which is what a module manufacturer can charge, the same two scenarios yield 37.8 and 23.1 million EUR per year; because material, energy, depreciation, financing, and logistics costs are not included, these figures are gross margins over labor cost and must not be read as net profits. The framework offers a transparent and reproducible way to compare production decisions across effectiveness levels; however, empirical validation requires plant-level data that are not yet available for Türkiye.

Keywords: solar energy; photovoltaic modules; employment; work; leisure time; economic output

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1. Introduction

The global energy landscape has witnessed a significant transition from conventional energy systems to renewable alternatives, primarily driven by the adverse environmental impacts of fossil-based energy production, such as climate change and greenhouse gas (GHG) emissions [1]. Among renewable energy sources, solar energy has emerged as one of the most promising options due to its sustainability and technological advancements. In particular, solar photovoltaic (PV) modules offer considerable economic and environmental benefits compared to conventional systems, including reduced costs, higher energy conversion efficiency, and a smaller ecological footprint ([2,3]).

Decision-making studies conducted for Türkiye have highlighted critical criteria for selecting PV technologies, such as technological maturity, conversion efficiency, GHG mitigation, cost-effectiveness, and a long service life [4]. PV systems significantly reduce emissions of CO_2 and other pollutants (e.g., SO_2, particulates), thereby minimizing negative impacts on human health and the environment through efficient utilization of limited natural resources.

For developing economies like Türkiye, domestic PV production represents a strategic approach to addressing the growing energy demand associated with population growth. Local manufacturing using domestic resources not only strengthens energy security but also contributes to economic resilience. This study compares production, employment and sales scenarios for a hypothetical PV module plant in Eskisehir under alternative labor-effectiveness assumptions. It does not assess the financial feasibility of such a plant, and no investment appraisal is carried out. Although several companies operate in the solar energy sector in Eskisehir, they are primarily EPC/solar installation companies that sell or install PV modules rather than PV module manufacturers. Because no PV module manufacturing plant currently operates in the city, the exercise is explicitly designed as a calibrated scenario analysis rather than an estimation based on observed plant-level data. The purpose of this scenario analysis is to make the assumptions and cost structure transparent and their implications reproducible, so that the framework can be re-run with plant-level data when such data become available.

1.1. Motivation: The Status of PV Modules in Türkiye

Investments in renewable energy are essential for improving welfare standards [5]. Solar energy, in particular, offers ecological benefits such as carbon emission reduction, efficient land use, and reliance on renewable resources [6]. Türkiye possesses favorable solar irradiance and geographic conditions; however, PV systems remain underutilized, contributing only 6% to national electricity generation and 11% to installed capacity in 2023 [7]. Although global solar energy generation increased by 23.2% in 2023 [8], further expansion of PV deployment is needed in Türkiye. Global PV capacity surpassed 1.7 TWp by 2025, underscoring the strategic importance of accelerating PV adoption in Türkiye [9].

Although Türkiye benefits from high solar potential and technical expertise [10], limited financial resources constrain domestic entrepreneurship relative to that in developed economies. Government incentives are expected to stimulate market demand, as evidenced by the cumulative installed PV capacity of 9.43 GW in 2022 [7]. Addressing the country's energy deficit through PV deployment requires both public and private investment under competitive market conditions. This study investigates the relationship between labor efficiency and human capital utilization in PV production, aiming to provide insights for sustainable energy policy.

Economic growth, energy consumption, and production factors—capital, labor, and human capital—are interdependent [11]. Previous research has explored these dynamics using physics-based analogies [12,13] and energy-inclusive production models [14,15],

yet gaps remain in explicitly integrating physical principles into economic modeling. This study addresses this gap by formulating a production function based on work-energy concepts, offering a novel perspective for analyzing decarbonization pathways in emerging economies.

1.2. Technical and Financial Debates for Strategic Planning Issues of PV Modules

Photovoltaic (PV) modules convert solar radiation directly into electricity and have become a cornerstone of global renewable energy strategies. Despite higher initial costs compared to conventional systems, PV technology offers significant benefits, including consumer savings, reduced transmission costs, and low emissions [16]. These advantages make PV systems a rational choice for Türkiye, provided that relevant financial considerations are adequately addressed.

Rapid commercialization positions PV modules as the foundation of future solar energy development [17]. Recent technological progress has shortened payback periods, improving economic feasibility. Beyond payback time, financial factors such as initial investment, annual PV electricity generation, and unit prices exert greater influence than subsidies or tax incentives. This study carries these financial considerations into a scenario comparison of PV manufacturing under competitive market conditions, without evaluating the financial feasibility of an investment.

1.3. The State of the Art: A Novel Economic and Financial Approach for a Firm in Eskisehir, Türkiye

This study conducts an economic and financial analysis of solar energy systems, focusing on PV module production and labor productivity in Eskisehir, Türkiye. Productivity is linked to human capital and R&D activities [18,19]. Primary data were collected through a local household electricity survey in 2013; these data describe potential residential demand and are not used to validate the production parameters. Under the stated assumptions, 15 production workers organized in three shifts and two support staff assemble 21.2 three-kilowatt module sets daily in the 100% effectiveness scenario. A plant producing 7738 sets per year, or 232,140 over a thirty-year horizon, would generate 1703 million EUR in revenue net of labor cost over thirty years. This corresponds to 56.8 million EUR per year at the installed-system price, or 37.8 million EUR per year when output is valued at the module share of that price. In both cases, these figures are calculated before material, energy, capital, financing and logistics costs, provided that demand is met. These magnitudes are scenario outputs conditional on the assumptions collected in Section 3.1; they are not evidence of the financial feasibility of an actual plant. The study introduces an expository representation of effective and non-utilized labor time within an output-based framework [12,13]. It is a representation and not a prediction, and it carries no claim about welfare or well-being.

To the best of our knowledge, this is among the first attempts to use the scalar-product representation from mechanics as an expository device for labor effectiveness in energy economics: $W = \vec{F} \cdot \vec{d} = Fd\cos\theta$. Here, $\cos\theta$ represents firm efficiency, adapted to economic terms. This formulation provides a practical tool for assessing labor efficiency and strategic production decisions under different scenarios. Additionally, PV manufacturing offers ecological benefits, including reduced emissions and cleaner energy, while creating employment opportunities.

The remainder of this paper is organized as follows: Section 2 reviews the current PV market and technical characteristics in Türkiye; Section 3 outlines the methodology; Section 4 presents empirical results and discussion; Section 5 concludes with policy implications.

2. Current Situation, Production Factors, and Technical Characteristics of PV Modules in Türkiye

2.1. The Current Outlook of PV Modules in Financial Terms and Its Market in Eskisehir, Türkiye, and in the World

The city of Eskisehir is in the northwestern part of Türkiye at 39° 47′ N, 30° 31′ E, with an area of 2678 sq km and an elevation of 788 m (2585 ft). Although it lies well above sea level, Eskisehir receives a considerable amount of solar irradiation. It has an annual bright-sunshine duration of 2479 h and an annual global solar irradiation of 1472 kWh/m^2. These two figures are different quantities and are not comparable with one another: the first counts the hours during which the sun is recorded as shining, whereas the second is an energy received per unit area. Dividing the second by the standard reference irradiance of 1 kW/m^2 converts it into a time, 1472 equivalent full-power hours per year, and it is this converted figure, not the 2479 h of sunshine duration, that is used in the calculations of this paper. Figure 1 shows the total solar radiation levels across the districts of Eskisehir. The colors in the bottom left correspond to each district's solar radiation calculations in kWh/m^2-year. The southern parts of Eskisehir, shown in yellow tones, are more suitable for utilizing solar radiation than other areas of Eskisehir.

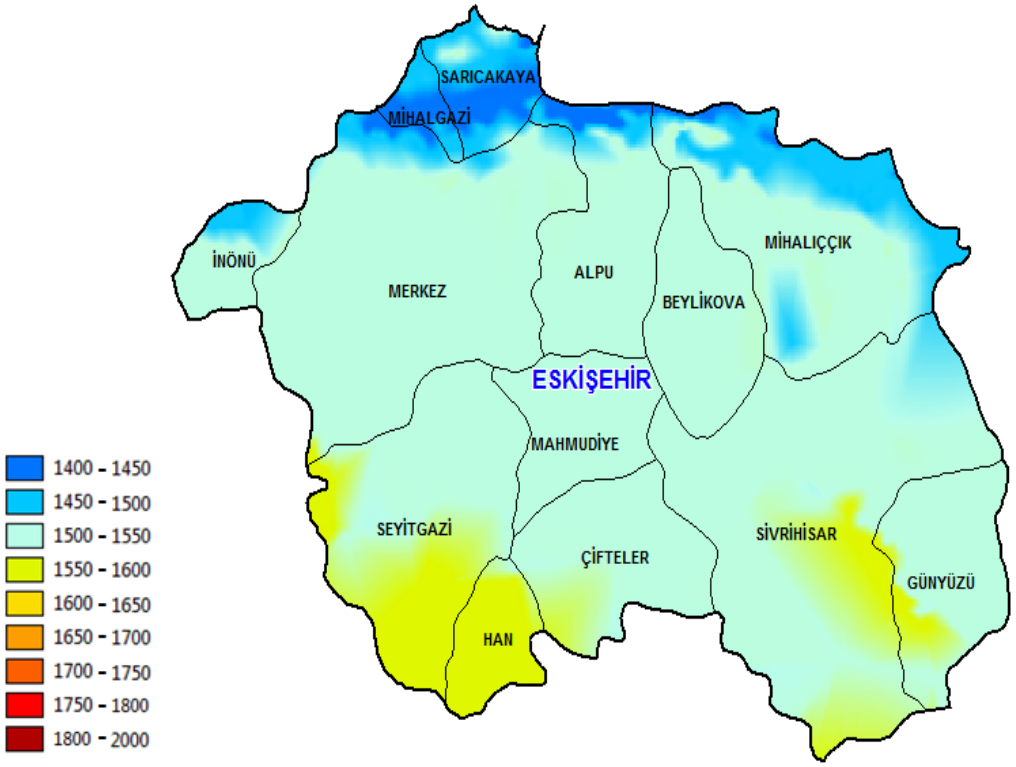


Figure 1. Solar energy potential map and total solar radiation level of Eskisehir by kWh/m^2-year. Source: [20]. The labels on the map are the names of the districts of Eskisehir Province; "MERKEZ" denotes the central district. The legend gives the annual global solar irradiation in kWh/m^2-year.

As shown in Figure 2, a PV module was installed on the right side of the roof. This picture is a complete depiction of the advantages of PV modules when they are utilized in Eskisehir. PV modules are environmentally friendly and have a minimal negative impact in terms of pollution. For those reasons, buying PV equipment can be a powerful and rational investment decision for consumers. Within this context, this study focuses on the use of domestic resources by examining the stages of self-production and assessing whether this investment strategy is rational over the specified time period. If it is rational, how plausible is it in financial terms?

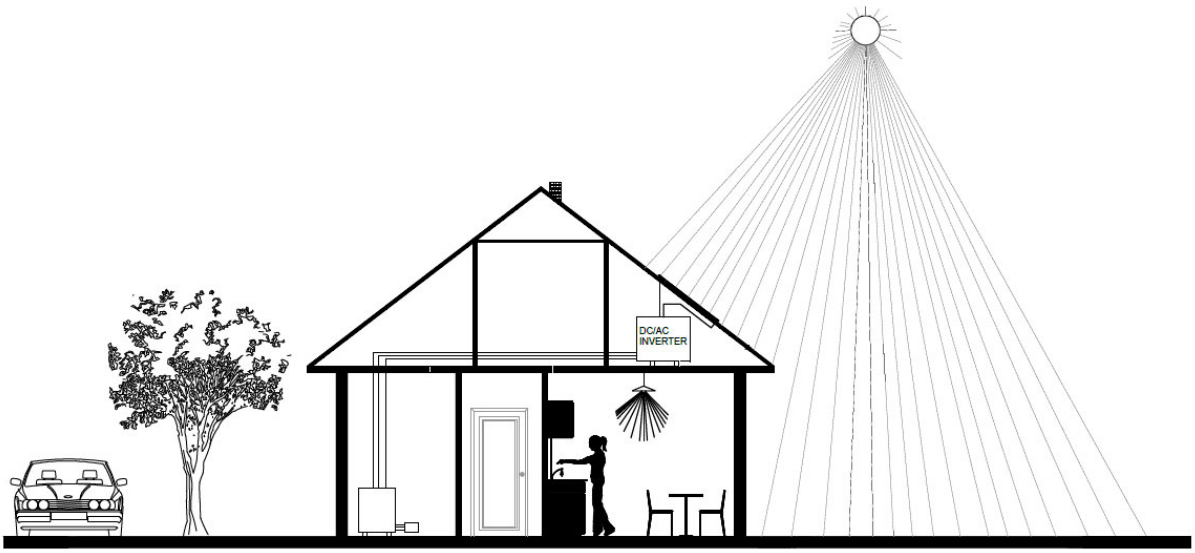


Figure 2. Scaled sectional drawing of a residential building illustrating the integration, utilization, and operating principle of a photovoltaic (PV) system. **Note:** This figure was originally created by the author specifically for this study using AutoCAD 2016. The residential building was drawn to scale to illustrate the integration of the PV system into the building and the flow of electrical energy from the PV modules to the household electrical loads. It is not reproduced or adapted from any previously published source.

Assuming that it is a rational decision, we set up a solar energy system. Consider an installed peak capacity of 1 kWp. One module of 180 Wp measures 1.58 m × 0.808 m = 1.28 m^2, so about 7.1 m^2 of module area is required per kWp (1000 Wp/180 Wp = 5.6 modules, 5.6 × 1.28 m^2 = 7.1 m^2); in practice, six modules, 7.7 m^2 and 1.08 kWp, are installed [21]. Throughout this paper, kW denotes power, kWp installed peak power, and kWh energy, and these units are kept distinct. However, the solar irradiance reaching the system varies throughout the year: in winter the daily irradiation falls, and some days are overcast. Two irradiation assumptions are therefore used in this paper and are kept apart throughout. Because irradiance is a power per unit area (kW/m^2) and irradiation is an energy per unit area (kWh/m^2), the conversion between them is written out once here and used in the same form wherever the two appear together. The irradiation-based assumption follows from the official atlas value by dividing an energy per unit area by a power per unit area, which leaves a time: (1472 kWh/m^2 per year)/(1 kW/m^2) = 1472 h per year, that is 1472/365 = 4.03, about 4.0 equivalent hours of full power per day. The illustrative upper case is 6 h per day, which by the same conversion in the opposite direction corresponds to 6 h/day × 365 days × 1 kW/m^2 = 2190 kWh/m^2 per year; it is a hypothetical variant and is not derived from the local irradiation data. Section 4.3.1 states the conversion and Appendix A Note 1 reports the payback calculation on both assumptions.

2.2. Domestic Manufacturing Factors Affecting PV Modules in Eskisehir and in Türkiye

2.2.1. The Effects of Human Capital and Efficiency of the Labor Force in a PV Module Firm

Innovation is a key factor for firms that are manufacturing technological goods. Today's economy relies heavily on the effective use of human capital as a factor of production, which in turn determines the firms' innovation capacity. One way to measure innovation capacity is through R&D activities over a long production period [18]. Innovation arises from the combination of human capital skills and R&D activities [19]. The firm considered in this study will also produce high-technology products, such as PV modules. Table A1 shows the relationship between the share of R&D personnel and the innovation capacity of defense and aviation firms in Ankara [22]. Therefore, it provides significant evidence for

the average human capital level of these firms in the related geography. Among firms with a high share of R&D personnel, 61.1% display high innovation capacity (Table A1). We state plainly what this number is and what it is not. It is a share within a categorical contingency table for defense and aviation firms in Ankara; it is not a measured production efficiency, labor productivity, or total factor productivity, and it does not refer to PV manufacturing or to Eskisehir. No plant-level productivity data exist for PV module manufacturing in Türkiye. In the absence of such data, we use 61.1% only as a benchmark value for the dimensionless effectiveness index H = cos θ, which is obtained from the trigonometric relation (i.e., arc cos(61.1/100)° → θ = 52.3°) of "arc cos(61.1/100)°", where θ is equal to 52.3°. The transformation into an angle is a change in notation, not an estimate: it adds no information beyond the assumed value, and every result conditional on it is therefore a scenario result. The opposite benchmark, H = 1 (θ = 0°), is the ideal case of fully effective labor. Section 4.4 reports how the results respond to changes in this parameter, and the value should be replaced with direct plant-level measurements whenever such data become available.

2.2.2. Production Factors Affecting the Manufacturing Process of PV Modules in a Firm

In this part, the following technical approach adopted in this study is introduced. The literature is carefully reviewed, and some adaptations are made to make it more realistic and to reflect practical conditions. For instance, we refer to the estimates in [23], as shown in Table A2 in the Appendix A. On the other hand, we used a different production function (i.e., a Cobb–Douglas function), which allows us to derive the maximization conditions with respect to the production factors. Since no PV module manufacturing firm operates in Eskisehir, no plant-level data are available from which output elasticities could be estimated, and the parameters are therefore calibrated rather than estimated. In the Cobb–Douglas specification, α, β, and γ are output elasticities with respect to energy, labor, and capital, and they must not be confused with elasticities of substitution. A Cobb–Douglas technology imposes a unit elasticity of substitution between every pair of inputs by construction, whereas $\alpha + \beta + \gamma = 1$ imposes constant returns to scale; these are different restrictions. Accordingly, the nested-CES substitution elasticities estimated for Germany [23] and reproduced in Table A2 are used only as a qualitative reference for the relative weight of the three inputs, and their numerical values are not transplanted into our production function. We set $\alpha = 0.2$ for energy, $\beta = 0.3$ for labor, and $\gamma = 0.5$ for capital, so that the three sum to unity and the capital-intensive character of module assembly is reflected in the largest share. These are the values used in every figure and every calculation reported below (Sections 3.1 and 4.2). Since the parameters are assumed rather than estimated, Section 4.4 reports how the scenario results change when they are varied.

2.3. Technical Characteristics of a PV Module

A solar cell is a p-n junction consisting of p-type and n-type semiconductors, as shown in Figure 3. Silicon, gallium arsenide, indium arsenide, and cadmium sulfide are widely used cell materials. Connecting solar cells in series forms a module that delivers the voltage required by the user. The conversion efficiency of a solar cell can be given by Equation (1).

$$\eta\% = \frac{P_{o(Electrical)}}{P_{i(lightenergy)}} \times 100\% = \frac{V_m I_m}{P_{in}} \times 100\% = \frac{P_{\max(device)}}{\left(100\frac{\mathrm{mW}}{\mathrm{cm}^2}\right)(Area(\mathrm{cm}^2))} \times 100\% \quad (1)$$

where Pm = VmIm and Pin are the maximum power output and the incident power, respectively [24,25].

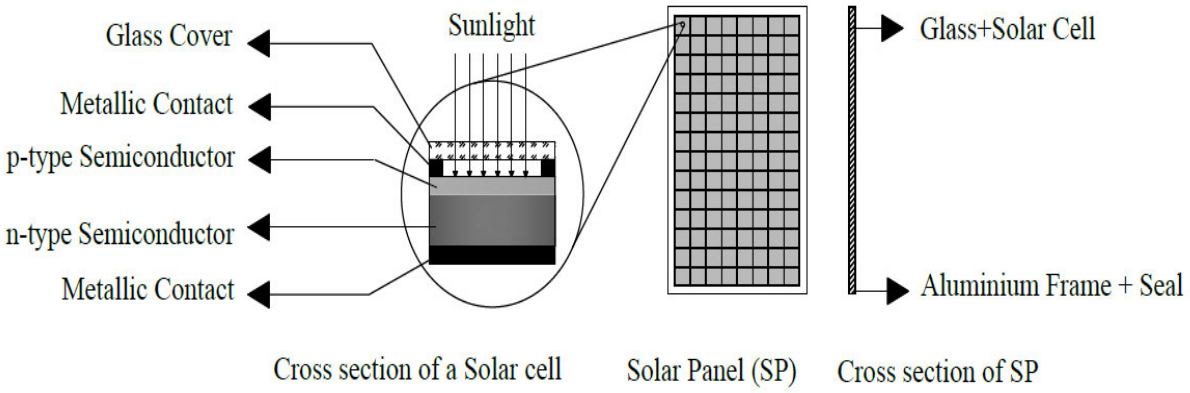


Figure 3. Technical characteristics and cross-sectional views of a photovoltaic (PV) module. **Note:** This figure was created by the author using AutoCAD 2016 to illustrate the structure and operating principle of a PV module.

The reported laboratory power conversion efficiencies of silicon, gallium arsenide, and CdS/CdTe solar cells are 24.1%, 28.8%, and 18.3%, respectively [26–28]. These are cell-level laboratory values; commercial module efficiencies are lower, typically 17–22%. The plant considered in this study is assumed to assemble crystalline-silicon modules of 180 Wp with an area of 1.28 m^2, corresponding to a module efficiency of about 14% at standard test conditions [26–28].

3. Research Method: Economics of PV Module Production

3.1. Chosen Production Function for a PV Module Firm

In this study, we specify a production function conveyed by the definition of work in physics, with the available technology level, considering the interactions of human capital with energy, labor, and capital. The economic model consists of input production factors to measure the possible production-cost structure of output and alternative employment opportunities in Eskisehir. The model's output is the number of three-kilowatt residential module sets assembled by the plant. This unit of output is used consistently throughout the paper, and Section 4.3.2 states explicitly which part of the installed-system price such a set can command. The inputs are the amount of energy consumed, the existing effect of human capital, capital expenses, the employed labor force, and the available technology level in a firm. A scholar used a production function that additionally included an energy input [15]. However, our approach is completely different because we applied the principles and laws of physics to our economic model. Our model is inspired by [12,13], but, again, we applied a completely different approach from their studies.

The main factors of production are energy, labor, capital, human capital, and technology. Therefore, an economic model can be set up around the referred variables, with their abbreviations: output (Y), energy (E), labor (L), capital (K), human capital (H), and technology (T). In this setup, E, K, H, and T are the inputs that are used to obtain Y. Y is the amount of PV modules produced by a firm. This relationship can be expressed in an economic model and is called a production function as follows in Equation (2):

$$Y(t) = f(E(t), L(t), K(t), H(t), T(t)) \quad (2)$$

To simplify the model, technology is treated as exogenous and assumed to grow exponentially at rate g. Since we analyze a single firm in one city rather than compare

firms or countries, we keep the technology specification as simple as possible. Accordingly, technology is specified as follows, where A is a technology constant in Equation (3):

$$T(t) = Aegt \quad (3)$$

A function is needed to express the model. Therefore, the Cobb–Douglas production function seems a suitable instrument for our purpose because terms such as "marginal product of labor" and "marginal cost" can be calculated by taking first-order derivatives of these types of functions. Real production can also be well approximated by it [29]. In addition to this, the law of "diminishing returns" in economics can be expressed by these kinds of functions. This law shows the characteristics of a firm's production and cost behavior and can serve as a guideline for producers or manufacturers. Regarding these considerations, by putting Equation (3) into Equation (2), the Cobb–Douglas production function takes the form of Equation (4) as follows:

$$Y(t) = Aegt \times (E(t)\alpha \times L(t)\beta \times K(t)\gamma) \times H(t) \quad (4)$$

Here, α, β, and γ are the output elasticities of Y(t) with respect to energy, labor, and capital; H(t) is exogenous, as is T(t). Because no time series of plant output exists, the technology term is a normalization and not an estimate: we set A = 1 and g = 0, so that T(t) = 1 over the scenario horizon and every difference between the reported scenarios stems from the effectiveness index H and from the input levels. Table 1 collects all parameter values, their sources, and their roles in the model.

Table 1. Model parameters, calibrated values, sources and roles.

A—technology constant = 1.	Normalization; no plant time series is available, so a is not estimated.
g—growth rate of technology = 0.	Static scenario horizon; technology is held constant across scenarios.
α—output elasticity of energy = 0.2.	Calibrated; constrained by $\alpha + \beta + \gamma = 1$.
β—output elasticity of labor = 0.3.	Calibrated; constrained by $\alpha + \beta + \gamma = 1$.
γ—output elasticity of capital = 0.5.	Calibrated; reflects the capital intensity of module assembly.
H = cos θ − effectiveness index = 0.611 (benchmark) and 1.000 (ideal).	Benchmark value assumed from the categorical share reported in [22]; see Section 2.2.1.
θ—effectiveness angle = 52.3° (benchmark) and 0° (ideal).	Re-notation of H; no additional information.
K—production scale parameter = 30, dimensionless, fixed in the short run.	Assumed, not observed. It is a scale index in Equation (4) and is not a monetary quantity: it is not the firm's share of the fixed cost, which is reported separately in euros in Section 4.2. Equation (11) applies only to long-run planning.
L—reference production workforce = 15 workers (assumed); output 21.2 sets per day (ideal) and 13.0 sets per day (benchmark).	Assumed. The output follows from Equation (4) at this workforce: with the scenario constraint E = L, Y = H times the square root of (K L), which gives 21.2 and 13.0 sets per day. The output index of Equation (4) is read one-to-one as three-kilowatt module sets per day; this reading is an assumption of scale and is not derived from engineering data. The effectiveness index scales output without changing the assumed scale, so both scenarios share the same workforce. The technical capacity limit of the installed plant is assumed to coincide with this scale.

Table 1. *Cont.*

P—price of one 3 kWp module set = 7350 EUR.	Installed-system price from [21]; an upper bound for a module manufacturer (see Sections 4.3.2 and 4.4).
w—labor cost = 500 EUR per employee per month (1 wage unit),	One unit being the Turkish minimum wage (500 EUR).
Monetary fixed cost = 250,000 EUR per year (500 wage units), that is 7,500,000 EUR over the thirty-year horizon	Under the subsidy scenario 235,000 EUR per year (470 wage units) is assumed to be subsidized and 15,000 EUR per year (30 wage units) is borne by the firm; under the market scenario the firm bears the full 250,000 EUR per year. This is a monetary cost and has no relation to the scale parameter K.
E = L—scenario constraint: the energy input is set equal to the labor input.	Imposed, not observed. It is a restriction of the scenario and is not derived from any equality of energy and labor prices. No separate energy price is identified in this study.
Input grid rule: labor is raised one worker at a time and energy follows it through the constraint E = L.	Simulation rule used to generate the production and cost curves presented in Section 4.2, according to our calculations.
Conversions, all on a common thirty-year basis: daily output (rounded to 0.1 sets) × 365 = annual output; annual output × 30 = horizon output; labor cost = employees × 500 EUR × 12 months × 30 years.	Applied in Section 4.3.2, according to our calculations.
Producer price variant = 4900 EUR per set, two-thirds of the installed-system price.	Module share attributable to the manufacturer; reported beside the system price in Sections 4.3.2 and 4.4.
Employees = 17: 15 production workers, organized in three shifts, and 2 support staff. Normalized cost weights: fixed weight F = 30 and a weight of 1 per production worker, with energy adding the same weight through E = L.	The cost weights are dimensionless indices used only for the cost curves presented in Section 4.2; they are not monthly or annual monetary costs, and every monetary magnitude is reported separately in Section 4.3.2.
Horizon = 30 years; operating days = 365 per year (assumption; maintenance shutdowns are not modeled).	
Module area = 7.1 m^2 per kWp; module rating = 180 Wp, module area = 1.28 m^2, equivalent full-power hours = 6 per day (illustrative upper case, equivalent to 2190 kWh/m^2 per year) or 4.03 per day (irradiation-based: 1472 kWh/m^2 per year divided by the reference irradiance of 1 kW/m^2, then by 365 days; see Section 4.3.1 and Appendix A Note 1), system losses = 10%.	

Note: Every value in this table is assumed or calibrated. None of them is an econometric estimate, and no standard errors are provided. These are the values applied in our calculations throughout the paper.

3.2. Profit Maximization of a PV Module Firm

The interval identifications for the parameters are as follows in Equation (5):

$$0 < \alpha, \beta, \gamma < 1 \quad (5)$$

Cobb–Douglas production function is predicted under the assumption of constant returns to scale, considering the energy, labor, and capital components, as represented in Equation (6):

$$\alpha + \beta + \gamma = 1 \quad (6)$$

Production and cost depend on the law of diminishing returns to varying inputs under a limited plant capacity. The model is solved in the short run: capital is fixed at the level implied by the initial set-up of the plant, only energy and labor vary, and the plant cannot exceed its installed capacity, so that Y is bounded above by Y-bar. In this short-run setting, the exponents of the two variable inputs sum to $\alpha + \beta = 0.5$, so the variable part of the

technology exhibits decreasing returns and, for given input and output prices, a finite interior optimum would exist. Constant returns hold over all three inputs together, that is, in the long run when capital also varies, and the argument about unlimited expansion under constant returns does not carry over to the short-run problem solved here. We nevertheless do not compute a profit maximum, because the price of energy relative to labor and the share of the installed-system price attributable to the manufacturer are assumptions rather than observations. The study is accordingly a comparison of cost scenarios: the operating points reported in Section 4.2 are the outputs at which average total cost is minimized for the installed plant, and no claim of profit maximization is attached to them. Here, the changing input or production factors are energy and labor. The domestic production of PV energy systems is a viable solution to addressing the unemployment problem in Eskisehir. Therefore, the workers will be employed to produce PV modules according to the marginal products of energy and labor, increasing the energy and labor inputs step by step. We set up a simulation that shows the firm's production characteristics as a function of the number of workers employed and the quantity of PV modules produced. In addition, we obtain the production and cost characteristics of the PV module firm, as presented in Section 4.2. In the following step, the entire cost family, which includes total costs, average costs, marginal costs, and their variable and fixed forms for a PV module firm, is drawn as output versus cost in the same section.

As a result of obtaining the cost structure of a PV module firm, the profit function can be defined as the difference between total revenue and total cost. The profit of a firm can be calculated as follows by choosing an appropriate production function. Our production function has three endogenous inputs: energy, labor, and capital, and two exogenous inputs: technology and human capital. The firm's profit-maximization conditions are presented in Equations (7)–(11):

$$\Pi = TR - TC \tag{7}$$

$$\Pi = p \times Y - c \times E - w \times L - r \times K \text{ the definition of profit} \tag{8}$$

$$\frac{\partial \pi}{\partial E} = 0 \text{ the maximization condition for energy} \tag{9}$$

$$\frac{\partial \pi}{\partial L} = 0 \text{ the maximization condition for labor} \tag{10}$$

$$\frac{\partial \pi}{\partial K} = 0 \text{ the maximization condition for capital} \tag{11}$$

In the above-mentioned equations, Π, TR, and TC are the symbols for profit, total revenue, and total cost, respectively (i.e., TR is the price equivalent of a PV module in a free market × output Y. The total cost is calculated as the sum of fixed capital cost and variable labor cost, including costs related to energy consumption.). In Equation (8), "p × Y" indicates the price times the output, which is equal to total revenue (TR); "c × E" represents the variable cost associated with energy consumption, "w × L" shows the variable cost incurred due to employed labor, and "r × K" displays the fixed cost stemming from the used capital. Y, E, L, and K denote the produced PV modules, consumed energy, employed labor, and paid or borrowed fixed capital inputs, respectively, as specified in Equation (4). The market prices of each item are identified by p, c, w, and r, as shown in Equation (8) [30]. Equations (9) and (10) are the short-run conditions for the two variable inputs and state that the value of the marginal product equals the input price; that is, p MP_E = c and p MP_L = w. Equation (11) is the corresponding long-run condition for capital, p MP_K = r; it is reported for completeness and is not imposed in the short-run scenarios of Section 4, in which capital is fixed. Note also that r in Equation (8) denotes the user cost of capital, whereas the growth rate of technology in Equation (3) is now denoted by g.

If we turn our attention to the convenience of the theory for practical life results, it can be added that life's purpose is not working for each possible working minute and maximizing our incomes. If this situation were true (or if it suited our purpose), no one would retire or go on a holiday. The more realistic purpose is the maximization of welfare, and this one is an example of an optimization. GDP informs us about how much physical output is produced in an economy, but it does not tell us how the physical gains are used for the optimization of happiness [31]. Starting from this point, in this study, the human capital input for Türkiye is used and is taken as exogenous for the calculation of a PV module firm's output. The human capital input that is summarized below (by a physical analogy), corresponding to "cos θ", is used within Türkiye's current specific labor force efficiency level for the production calculations. This analogy is built and corresponds to scalar multiplication in physics, and as far as we know, it is first used in [12] (but without mentioning "cos θ").

3.3. An Analogy from Physics: Expressing the Production Function in Terms of Work and Leisure Time Using the Scalar Product

The most important contribution of this study to the literature is as follows. Here, we transform a very well-known physics law "$W = \vec{F} \cdot \vec{d}$" to economics. An analogy is made for an appropriate expression of H(t) in the Cobb–Douglas production function. A production function can be expressed in the form of "Work" (at the same time in the form of "Leisure Time"). In Figure 4, this analogy is explained in an abstract display.

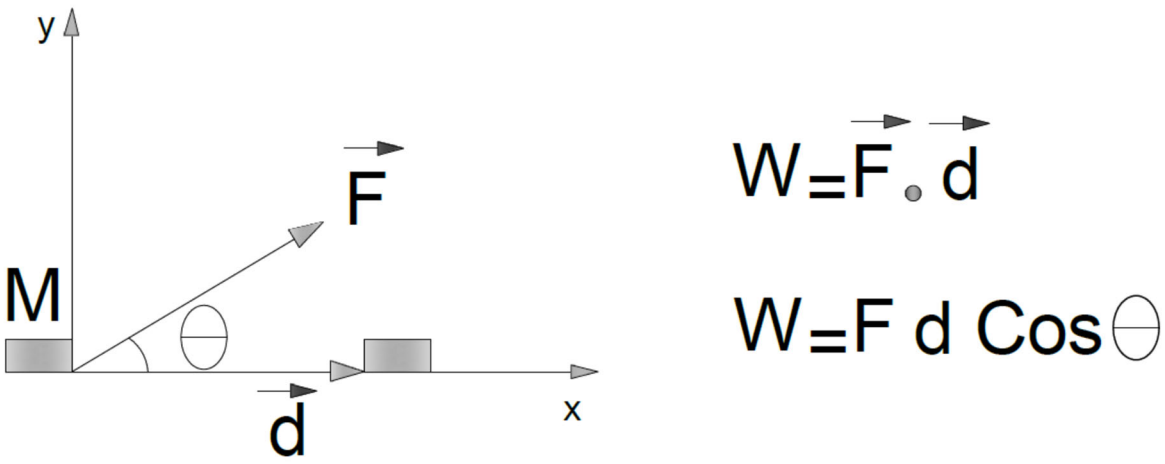


Figure 4. An analogy from physics: Mechanical work performed by a force acting on an object over a displacement. **Note 1:** This figure was created by the author specifically for this study to illustrate the concept of mechanical work and to demonstrate how a fundamental principle of physics can be applied in an economic context. **Note 2:** Cos θ takes values between zero and one. According to our analogy, the maximum efficiency of labor is equal to one. Therefore, if the θ angle approaches zero, the efficiency of labor increases and the work increases accordingly. On the other hand, if the θ angle approaches 90°, the work decreases. At the angle of 90°, the efficiency of labor equals zero. In other words, there is no effective work. We refer to this limiting case as the zero-effectiveness state. We avoid the term strike, which would attribute a single industrial-relations cause to an outcome that can arise for many reasons.

Here, if a force (F) of 90° is applied to a load M, this means that the applied F is perpendicular to the X axis, or that θ, which is the angle between the applied F and the X axis, is 90°. If F is applied along the X axis, that means that the angle between F and the X axis is zero. Thus, by analogy, the employed workers achieved the highest performance, and for this reason, they have the highest human capital value (i.e., the workers have appropriate human capital skills), while the relationship between consumed energy and

production factors operates in resonance. In other words, the maximum production can be supplied under this condition. In addition, if F is applied to a load at an angle of 45°, the effect of F is F × cos θ (θ = 45°) ≈ 0.707 F. In this case, we understand that there is a mismatch between the motivation of workers (the workers have missing human capital skills) and the firm's aim of profit maximization. If θ equals 90°, effectiveness is zero, and no effective work is performed, whatever the underlying cause: skill mismatch, organizational failure, equipment breakdown, or industrial action. The firm faces a short-term shutdown with no production. Table A3 in the Appendix A displays the various productivity combinations. Productivity is maximum at θ = 0° and minimum at θ = 90°, the zero-effectiveness state, consistent with Table A3.

With the above-mentioned assumptions, "Work" can be obtained from two vectors, the Force ($\vec{F}$) and the displacement ($\vec{d}$), as it is used in physics in Equation (12):

$$W = \vec{F} \cdot \vec{d} \quad (12)$$

Here in Equation (12), "$\vec{d}$" represents displacement. In scalar form, Equation (12) can be written as Equation (13):

$$W = F \times d \times \cos\theta \quad (13)$$

Here, d is the distance, that is, the magnitude of displacement. With the implementation of the laws of physics in economics, we obtain a relationship between "Leisure time" and "Work" that is controlled by efficiency and is displayed in normalized, dimensionless form, as shown in Figure 5a and as set out after Equation (14).

Our analogy is that "F" corresponds to "E", "d" corresponds to "(L × K)", and cos θ corresponds to "H". In Figure 5b, one can concentrate on the θ value of 52.3°, which refers to the 61.1% efficiency of a technological firm in Ankara, as we have already mentioned in Section 2.2.1.

Since there are possibilities that depend on efficiency and productivity differences in various firms, we can theoretically explain this situation by taking the geometrical place of all the possible conditions, as stated in Figure 5c, with dots.

If we increase the angles when those values are carried by the graph, the front of the arrows forms a quarter circle. Writing the two axes in the normalized form defined below, with w(t) = Work(t)/R and s(t) = Leisure(t)/R, the curve is characterized by Equation (14):

$$(w(t))^2 + (s(t))^2 = 1,\ 0^\circ \leq \theta \leq 90^\circ \quad (14)$$

Equation (14) is stated above in a normalized, dimensionless form. Let R = E (L K) denote the geometric scale of the construction, that is the radius of the quarter circle, and define w(t) = Work(t)/R = cos θ and s(t) = Leisure(t)/R = sin(θ). R must be distinguished from the output Y of Equation (4): Y is the plant output, measured in three-kilowatt module sets per day, whereas R is only the scale by which both axes of Figure 5a–d are divided. The two are different quantities and neither is computed from the other. Then w(t) squared plus s(t) squared equals one identically, which is what the quarter circle of Figure 5a–d displays. Two consequences follow and are stated explicitly. First, the relation is an identity by construction and carries no testable content: it cannot be rejected by any data, and it is not a derivation of labor supply. Second, once both axes are normalized by the same scale factor, both sides of the equation are commensurable; measuring Work and Leisure in hours while measuring Y in physical units of output would not be. What the construction provides is a compact representation of a single number, the effectiveness index cos θ, along with its complement. A genuine labor–leisure trade-off would require a time constraint, a utility function, a wage, and the resulting income and substitution effects, none of which

are modeled here. For this reason, s(t) is described throughout this paper as non-utilized labor time rather than as welfare-relevant leisure, and the construction is presented as an expository device. This continuous form is shown in Figure 5d.

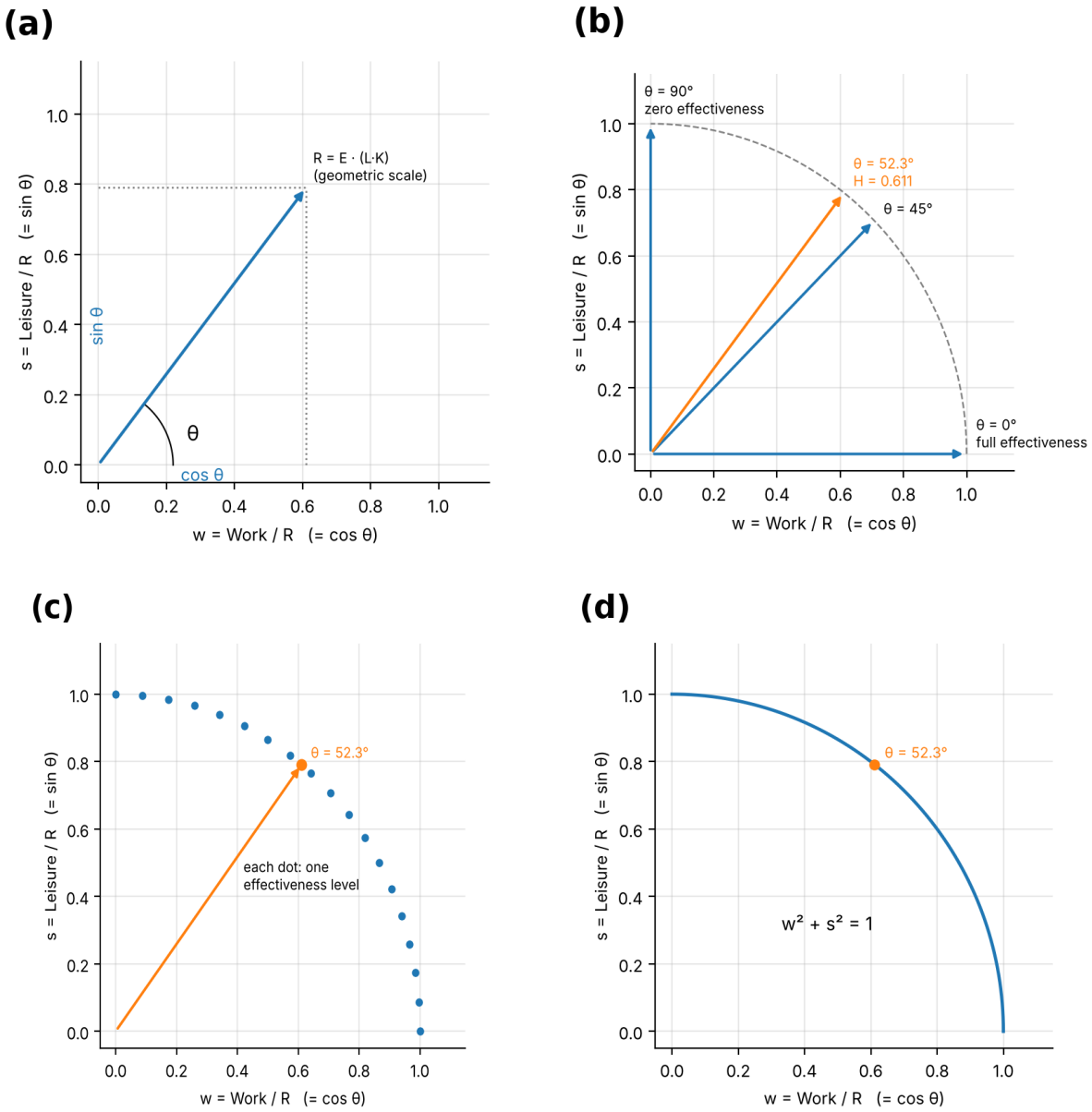


Figure 5. (**a**) An explanation of physics concepts in terms of labor economics: "Leisure time" versus "Work" controlled by efficiency, in normalized, dimensionless form (both axes divided by the geometric scale R, see Equation (14)). The horizontal axis is w = Work/R = cos θ and the vertical axis is s = Leisure/R = sin θ; neither axis is measured in hours, and R is not the output Y of Equation (4). (**b**) "Leisure time" versus "Work" in normalized, dimensionless form: some critical points, including θ = 52.3°, which corresponds to the benchmark effectiveness H = 0.611. Horizontal axis w = Work/R, vertical axis s = Leisure/R. In this panel the orange arrow marks the benchmark case θ = 52.3°, which corresponds to H = 0.611, and the blue arrows mark the other critical angles. (**c**) "Leisure time" versus "Work" in normalized, dimensionless form: each possible critical point, including θ = 52.3°. Horizontal axis w = Work/R, vertical axis s = Leisure/R. (**d**) "Leisure time" versus "Work" in normalized, dimensionless form: continuous form, w squared plus s squared equal to one. Horizontal axis w = Work/R, vertical axis s = Leisure/R.

3.4. The Meaning of W = F d cos θ in Energy Economics

The economic content of the analogy is stated here directly rather than left to the figures. In mechanics, only the component of a force that lies along the direction of displacement performs work; the perpendicular component is present, but it does not contribute. The economic reading adopted here is the same in form. A plant engages inputs—energy, labor, and capital—but only the part of that engagement which is aligned with the production objective is converted into output. The index cos θ measures that alignment: cos θ = 1 means that all engaged input capacity is converted into output, cos θ = 0 means that none of it is, and intermediate values represent partial alignment arising from skill mismatch, organizational friction, interruptions, or equipment problems. The complementary term sin(θ) represents engaged capacity that is not converted into output.

Two limits of this reading must be stated with equal clarity. First, energy is not a force and the product of labor and capital is not a displacement; they have neither the dimensions nor the direction of the corresponding physical quantities, and no physical angle can be defined between them. The construction is therefore an analogy used for exposition and normalization, not a derivation, and nothing in the empirical results depends on the physical interpretation. Second, cos θ enters the production function exactly as a Hicks-neutral effectiveness factor would: Y is proportional to H. The value added by the representation is expository—it displays effectiveness and its complement in a single normalized diagram—and not predictive.

3.5. Model System, Parameters and Computational Procedure

The computational procedure is as follows, and every magnitude reported in this paper follows from it step by step. (i) Fix the technology normalization A = 1 and g = 0 and the output elasticities $\alpha = 0.2$, $\beta = 0.3$, $\gamma = 0.5$. (ii) Fix the effectiveness index at H = 0.611 for the benchmark scenario and H = 1 for the ideal scenario. (iii) Hold the scale parameter K fixed and increase labor stepwise, with energy following it through the scenario constraint E = L, computing output from Equation (4) at each step. (iv) Compute the normalized cost family at each output level from the fixed weight F = 30 and the weight of 1 per production worker, so that the total weight is F + 2L; these weights are dimensionless indices and are not monetary costs. (v) Take the reference workforce L = 15 as assumed in Table 1 and read the corresponding output from Equation (4): with E = L the function gives Y = H times the square root of (K L), that is 21.2 module sets per day in the ideal case and 13.0 in the benchmark case. The output index is read one-to-one as module sets per day, which is an assumption of scale. Neither the workforce nor the capacity is obtained as an economic cost optimum; at the chosen weights, the normalized average cost curve happens to attain its minimum at this same reference scale, so the weight F and the reference workforce are two expressions of one assumption rather than independent results. (vi) Scale these daily figures to annual and thirty-year output. (vii) Value output at the price of one module set and subtract labor cost to obtain the gross margin reported in Section 4.3.2. Our calculations follow the same order: the parameter block of Table 1, including the scale parameter K and the reference workforce L; the input grid, in which labor is raised one worker at a time and energy follows through E = L; the normalized output and cost columns that generate the production and cost curves presented in Section 4.2, with marginal cost computed analytically as the derivative of total weight with respect to output rather than as a difference between adjacent rows; and the conversion block, in which daily output is first rounded to 0.1 sets and every monetary magnitude in this paper then follows from the rounded value on a common thirty-year basis. Each assumption is identified in Table 1 and each step above states what is computed from it, so that the daily output, the cost illustration and the financial results can be recomputed independently.

In the theoretical part, we have demonstrated a firm's objective, which is profit maximization under free-market conditions. In the following sections, we explain how the theoretical model works in practice. Before proceeding, we examine the data required for the economic model.

4. Data, Results, and Discussion

4.1. Data, Survey Questions, and Answers

The data were collected firsthand through a survey conducted in 2013. The survey was conducted in the city of Eskisehir, and the respondents were residents of Eskisehir. The questionnaires were distributed in person to participating residents. We first determined the number of households and the average number of members in a nuclear family in the sample; this provided a general overview of the surveyed population. We accounted for seasonal variation by collecting separate responses for each month. Our aim was to measure the total energy consumption of each household. The survey was designed for multiple purposes. However, this study concentrates on household electricity consumption. This is because electricity consumption is the intersection point between PV module use for electricity generation and households' energy demand. In addition, residents were asked about the type of house they lived in and the electrical devices they had. In line with these questions, respondents were asked about energy use for heating, cooling, and lighting in the residence.

The questionnaire combined open-ended and multiple-choice items. The sampling frame was households in the central districts of Eskisehir; questionnaires were delivered by hand to households selected without a formal probability design, so the sample is a convenience sample and is not claimed to be representative of the city. A total of 154 questionnaires were collected and numbered. Nine of them, respondents 21, 25, 26, 48, 49, 71, 111, 133 and 138, were removed because their monthly consumption entries were incomplete, so the final dataset supplied as Supplementary Materials contains 145 records (Survey-Data.xlsx, English sheet, rows 2 to 146), and these 145 records are the observations analyzed here. The gaps in the respondent numbering of the file correspond exactly to the nine removed questionnaires, so the three counts—collected, removed, and analyzed—can be checked directly against the dataset. No other count of respondents appears in the paper. The questionnaire contains no item on household income or occupation (Table A4), so no claim of socio-economic representativeness is made. The number of valid answers differs across items and is now reported item by item in the note to Table A5: the monthly figures expressed in kWh rest on 8 respondents for January to November and 9 for December, whereas the figures expressed in TL rest on about 142 respondents, because most households reported their bills in currency rather than in kWh. This difference is a limitation of the data and is discussed in Section 4.4. Item 4 of the questionnaire refers to electricity used for lighting, and it is treated here as a lower-bound indicator of household electricity use rather than as total household consumption. The data were collected firsthand by the authors. Table A4 in the Appendix A shows the survey questions, and Table A5 presents more detailed statistics on the corresponding responses.

Descriptive statistics from the same 2013 survey were reported earlier in [32]. The overlap is confined to the monthly descriptive means of questionnaire items 4 and 6.1 reproduced in Table A5; the production and cost model, the effectiveness parameterization, the scenario design, Table 1, the scenario comparison presented in Section 4.4, and every financial magnitude reported in Sections 3 and 4 are new to this paper. The institutional framework in force at the time of the 2013 data collection has been forwarded to the editorial office. The survey was conducted in 2013 at Eskisehir Osmangazi University. Participation was voluntary and anonymous; no personal identifiers were recorded, and oral informed

consent was obtained before each questionnaire was completed; the corresponding statements are given after the Conclusion. The complete survey dataset is provided as part of the Supplementary Materials.

Based on the response to the initial question, we determine that an average-sized family unit in Eskisehir has three members. The annual population growth rate used consistently in this paper is 2.2% [33], which is the value applied in the projections in Section 4.3.2.

4.2. Results: Applying Economic Theory to Practical Experience

Changes in output relative to changes in total cost are examined to better understand the production-cost behavior of the PV module firm. In addition, the relationship between output and total costs determines the cost function of a PV module firm in Eskisehir based on the available data. The cost function is obtained from the geometric relationship between total cost and the quantity of output. Representing the production factors in units simplifies the derivation of the production and cost functions and the calculation of the cost structure of the PV module firm. A single unit is defined as the equivalent of the minimum wage in Türkiye. The variable cost rises as additional workers are employed, and the amount of energy consumed equals the sum of their energy consumption for a given level of production.

If the building and equipment costs are considered, approximately 7,500,000 EUR of fixed cost can be assumed for the thirty-year period, that is 15,000 wage units at a minimum wage of 500 EUR per month. The annual monetary fixed cost is therefore 250,000 EUR, that is 500 wage units per year. This is a monetary magnitude on an annual basis and is unrelated to the scale parameter K of Equation (4), which is a dimensionless index. Since the Turkish government gives priority to renewable energy investments, we propose and anticipate that it will subsidize a major part of the fixed capital amount to encourage domestic investors. This kind of investment has many strategic advantages for a country's development, such as creating employment opportunities, increasing energy independence, and contributing to a clean environment and, more broadly, to sustainable development [34]. In addition, the firm will quickly pay back this subsidy amount once production begins, so only a minimal portion of the fixed cost, 15,000 EUR per year (30 wage units), is borne by the firm under this scenario. The remaining 235,000 EUR per year (470 wage units) is assumed to be subsidized by the government as part of the initial investment strategy. Under the market scenario the firm bears the full 250,000 EUR per year.

The production and cost functions of a PV module firm are shown in Figure 6a,b. In this study, a high-efficiency working principle is explained by the efficient use of the labor force or the effective use of human capital. The benchmark scenario corresponds to the 61.1% value taken from [22], where Koroglu and Eceral report firm-level evidence for the Turkish high-technology industry, and the ideal scenario is the fully effective case H = 1. The labeling of the two scenarios is uniform throughout this paper: 61.1% is always the benchmark or low-effectiveness case, and 100% is always the ideal or high-effectiveness case. As explained in Section 2.2.1, the benchmark value is an assumption and not a measurement of PV manufacturing efficiency. The total amount of output increases at a decreasing rate, which exactly reflects the main characteristics of a production function in both the high-efficiency and low-efficiency modes, represented by the solid and dashed lines, respectively, in Figure 6a. In addition to this, the total cost increases at an accelerating pace in line with the increasing quantity of output, which also fits the classical cost function pattern, as indicated in Figure 6b. The law of diminishing returns explains the simultaneous relationships between the production and cost functions of a PV module firm in Eskisehir.

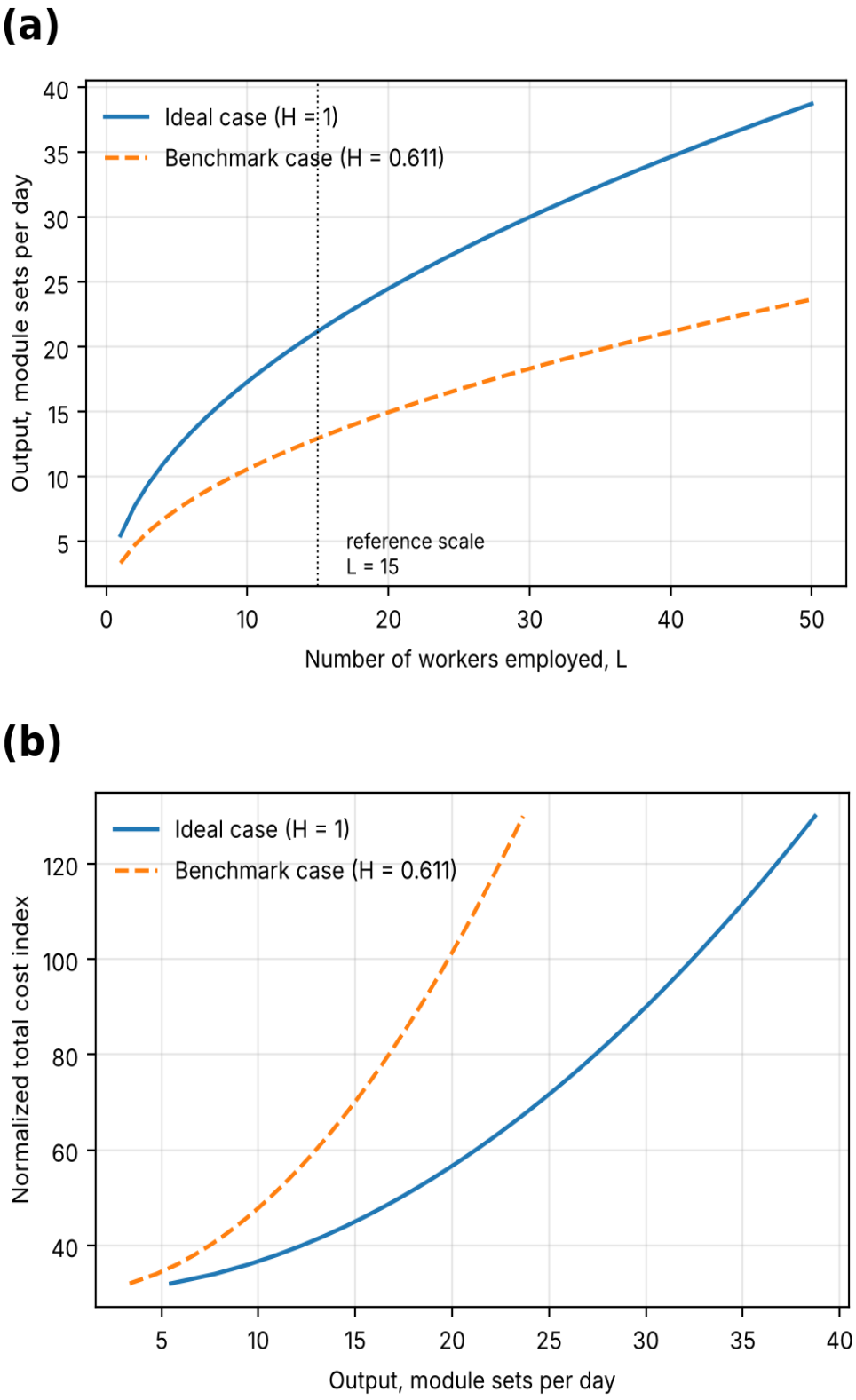


Figure 6. (**a**) Production function of the plant under ideal (H = 1) and benchmark (H = 0.611) effectiveness. Horizontal axis: number of workers employed, L; vertical axis: output in module sets per day. The assumed reference scale L = 15 is marked. Plotted from our calculations. (**b**) Cost function of the plant under ideal (H = 1) and benchmark (H = 0.611) effectiveness. Horizontal axis: output in module sets per day; vertical axis: normalized total cost index (fixed weight F = 30 plus a weight of 1 per production worker, with energy entering through E = L). The index is dimensionless and is not a monthly or annual monetary cost. Plotted from our calculations.

One can find the values of average (fixed, variable, total, and marginal) costs. These values are shown in Figure 7a,b as cost family curves of a PV module firm under high- and low-efficiency scenarios, respectively (i.e., to determine the amount of output, we use a Cobb–Douglas production function that has the values A = 1 (because technology is treated as exogenous and no evaluation or comparison across countries is conducted) and $\alpha + \beta + \gamma = 1$ because of constant returns to scale, and the parameters are specified as follows: E = 0.2, K = 0.5, and L = 0.3). Marginal cost is computed analytically as the derivative of the total weight with respect to output, not as a difference between adjacent rows of the input grid. It rises monotonically with output in this specification because the two variable inputs are increased together; it lies below the average total cost curve while that curve is falling and meets it at an index value of 2.83 in the ideal case and 4.63 in the benchmark case. The attributes of the marginal cost and the increasing number of employed workers determine the structure of average variable cost. This is because, when the number of employed workers increases, the output increases, and this rise is adjusted by the marginal product of labor. In addition, the average fixed cost is determined from the ratio of fixed cost to the output quantity and decreases as output increases, as shown in Figure 7a,b. Since the variable cost increases at a decreasing rate, the output amount simultaneously increases at a decreasing rate. The summation of the average fixed and variable costs is called the average total cost. Three things are kept apart at this point, and the paper claims no more than the third. First, the reference workforce of 15 production workers is an assumption of plant scale, stated in Table 1; it is not obtained from any economic optimization. Second, the output that follows from it is read from Equation (4): with the scenario constraint E = L the function gives Y = H times the square root of (K L), that is 21.2 module sets per day in the ideal case and 13.0 in the benchmark case, the index being read one-to-one as sets per day. The effectiveness index scales output without changing the assumed scale, so the two scenarios share the same workforce and differ only in output. Third, Figure 7a,b are a normalized illustration of the shape of the cost structure. Their weights are a fixed weight F = 30 and a weight of 1 per production worker, with energy entering at the same weight through E = L, so that the total weight is F + 2L. These weights are dimensionless indices: they are not monthly or annual monetary costs, and no monetary conclusion is drawn from them. With F = 30 the normalized average cost curve attains its minimum at the same L = 15, so the weight and the reference workforce are two expressions of a single assumption rather than two independent findings. The technical capacity limit of the installed plant is a further restriction, assumed here to coincide with this scale. All monetary magnitudes are reported separately in Section 4.3.2, where costs and the corresponding output are placed on a common thirty-year basis.

4.3. Discussion

4.3.1. The Convenience of the Economic Theory with the Survey Data: A Probable Scenario About Utilization of PV Modules

The long-term basis for the irradiation assumption is the official solar atlas value for Eskisehir, 1472 kWh per m^2 per year [20], reported in Section 2.1. As a corroborating spot measurement, one of the authors measured solar irradiance in Eskisehir on 15 July 1986 between 10:00 and 16:00 and obtained an average irradiance of 546 W/m^2 over that six-hour interval [35]. This measurement cannot be set against the atlas value as it stands, because 546 W/m^2 is a power per unit area while the atlas reports an energy per unit area; multiplying by the length of the interval brings both to the same unit: 546 $W/m^2 \times 6$ h = 3276 Wh/m^2 = 3.28 kWh/m^2 received in that window. The atlas value corresponds to a daily average of 1472/365 = 4.03 kWh/m^2, so the six-hour window captured about 81% of an average day, which is the expected order of magnitude for a window that excludes the early morning and the late afternoon. A single-day measurement cannot support a

thirty-year production profile; it is reported only as an order-of-magnitude check, and the limitations arising from irradiation variability, tilt, orientation, temperature, and system losses are set out in Section 4.4 [35]. The two irradiation assumptions used in this paper are stated together here. The atlas value of 1472 kWh per m^2 per year, divided by the standard reference irradiance of 1 kW per m^2, gives 1472 equivalent full-power hours per year, that is 1472/365 = 4.03, about 4.0 h per day. The figure of six equivalent full-power hours per day used in Appendix A Note 1 corresponds instead to $6 \times 365 \times 1$ kW/m^2 = 2190 kWh per m^2 per year and is therefore an illustrative upper case rather than a value derived from the local irradiation data. Appendix A Note 1 now reports the payback calculation on both assumptions.

Five quantities enter the sizing of a residential system, and they are labeled here so that none of them is compared with another before both are expressed in the same unit. (a) is power, measured in kW: the rate at which appliances draw electricity at a given instant. (b) is energy, measured in kWh, consumed over a stated period, here one day. (c) is the daily electricity use implied by the survey, in kWh per day, on the same unit as (b). (d) is the installed peak capacity of the system, in kWp. (e) is the module area it requires, in m^2. Power becomes energy only after multiplication by a duration, and energy becomes power only after division by one; nowhere in what follows are the two added together or set against each other directly.

Taking (a) first, the nameplate powers of the appliance inventory are as follows. Ten light bulbs of 60 W each draw 10×60 W = 600 W = 0.6 kW; a refrigerator draws about 0.3 kW; a television draws between 0.3 and 0.6 kW. These three are present more or less continuously and together draw about 1.2 to 1.5 kW. One large appliance in use at the same time, a water heater, an oven or a flat iron, adds roughly 1.5 to 2 kW. The peak simultaneous load of the inventory is therefore (a) = about 3 kW. This figure is power and says nothing by itself about consumption over a day.

Turning to (b), the appliances that run in cycles rather than continuously are reported per cycle: a flat iron and a water heater consume about 3 kWh each, a washing machine and a dishwasher between 1.5 and 2 kWh each during the heating of the water, and an oven about 2 kWh. These are energies per cycle, not per day, and a household does not run every cycle every day. Depending on how many cycles fall on a given day, the daily electricity consumption of a residence is taken here to be (b) = between 3 and 6 kWh per day, 3 kWh at the lower end and 6 kWh at the upper end. This range is an assumption about how often the cycles occur; it is not the sum of the list above, which would be larger.

The survey supplies (c) on the same unit as (b). The monthly mean of item 4 in Table A5 is 142.4 kWh, that is 142.4/30 = 4.7 kWh per day. Item 4 is worded as electricity used for lighting, so (c) is a lower bound for total household electricity use rather than a measurement of it. Read strictly as lighting, 4.7 kWh per day against a lighting load of 0.6 kW implies 4.7/0.6 = 7.8 h per day with all ten bulbs on, which is high; it is at least as likely that respondents reported a broader part of their bill. Under either reading (c) falls inside the range assumed for (b), and (c) is the figure carried into the monetary calculations of Section 4.3.2, where it is treated throughout as a lower bound.

The system is then sized. Matching the peak load (a) gives (d) = 3 kWp and, at 7.1 m^2 per kWp from Section 2.1, (e) = 3×7.1 = 21.3 m^2. This is a conservative, peak-based sizing rule, and we adopt it explicitly. Sizing on daily energy instead converts (b) into a power by dividing it by the equivalent full-power hours and by the loss factor: 3 kWh per day/(6 h per day $\times$ 0.9) = 0.56 kWp on the illustrative six-hour assumption, or 3 kWh per day/(4.03 h per day $\times$ 0.9) = 0.83 kWp on the irradiation-based assumption, giving about 4 m^2 and about 6 m^2 of module area respectively. Energy-based sizing therefore yields a system about five times smaller than the peak-based one on the illustrative assumption and

about four times smaller on the irradiation-based one, and it would reduce the investment per household, the market volume and the revenue figures of Section 4.3.2 in the same proportion. This sensitivity is reported in Section 4.4 (see Appendix A, Note 2 for the cost scenario).

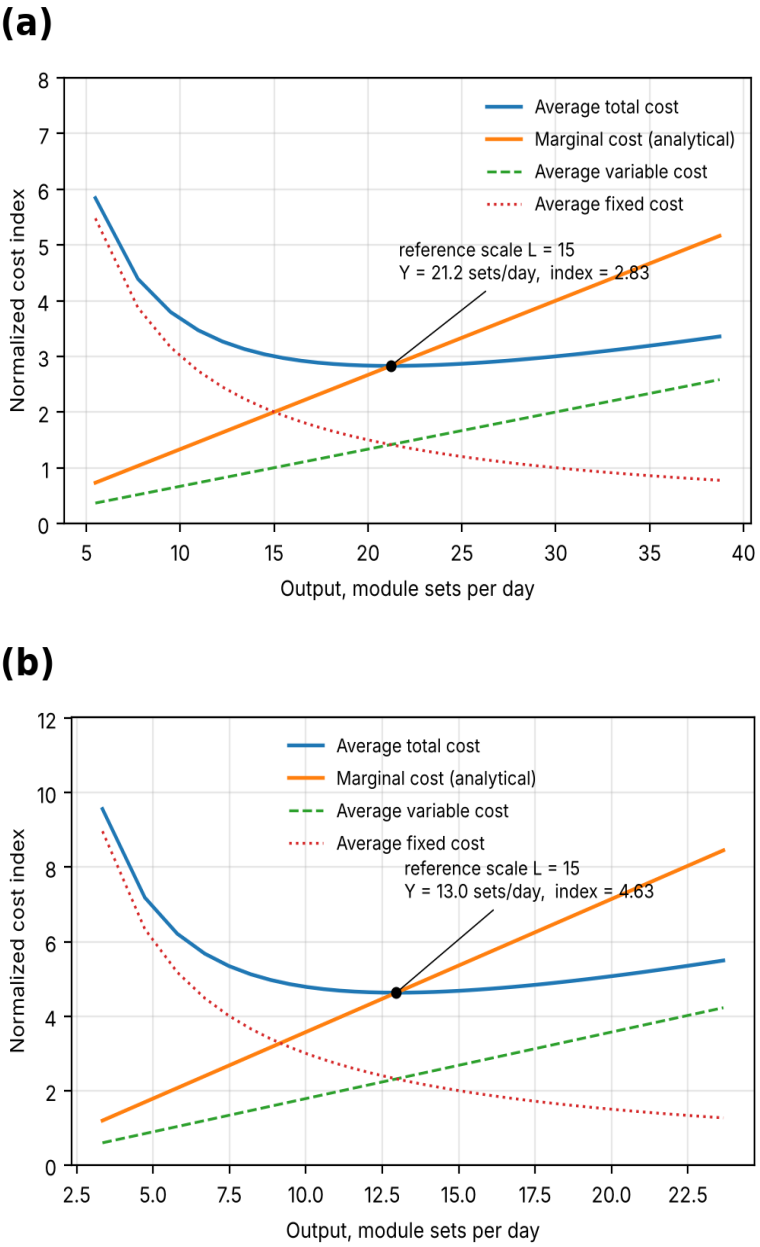


Figure 7. (**a**) Cost family in the ideal case (H = 1), with output elasticities α(E) = 0.2, β(L) = 0.3 and γ(K) = 0.5. Vertical axis: normalized cost index, dimensionless, from the fixed weight F = 30 and a weight of 1 per production worker. Marginal cost is computed analytically and meets average total cost at the assumed reference scale L = 15, that is at 21.2 module sets per day and an index value of 2.83. This coincidence follows from the chosen weight F and is not a derived economic optimum. Plotted from our calculations. (**b**) Cost family in the benchmark case (H = 0.611), with output elasticities α(E) = 0.2, β(L) = 0.3 and γ(K) = 0.5. Vertical axis: normalized cost index, dimensionless, from the same weights as in Figure 7a. Marginal cost is computed analytically and meets average total cost at the assumed reference scale L = 15, that is at 13.0 module sets per day and an index value of 4.63. The workforce is the same as in the ideal case, since the effectiveness index scales output without changing the assumed scale, and the coincidence at that scale follows from the chosen weight rather than from an economic optimum. Plotted from our calculations.

There is an important technical issue that needs to be discussed. The electricity generated by a PV module needs to be converted to an appropriate voltage level by an inverter. For example, for the refrigerator and light bulbs, the electricity needs to be converted to 24 volts (V). On the other hand, by converting the electricity to 220 V alternating current, energy losses will be minimized, and this process is a necessary preference for consumers for energy savings. Manufacturers typically warrant that a module still delivers at least 80% of its nameplate power after 20 years. This is power retention under degradation, not conversion efficiency; the two are distinct quantities. Degradation is not modeled in the financial scenarios below. Some firms, such as General Electric, provide the indicated warranty for the duration of use.

4.3.2. The Convenience of the Economic Theory with the Practical Life: Revenue, Labor Cost and Gross Margin Calculations Using the Survey Data

In this section, we compute revenue and labor costs for the plant and report the resulting gross margin. The survey data are used for the calculations. We sought consistency between the theoretical estimates and the survey data. For the sake of realism, we used "kWh" instead of "TL" as the unit of measurement. Although both units were available in the survey, the use of "TL" could possibly be affected by inflation rates. The average electricity use for lighting in a residence, as indicated in Table A5, was 142.4, that is, about 142 kWh per month. Table A5 reports the same item in currency as 60.58 TL per month, so the implied residential tariff is 60.58/142.4 = 0.425 TL per kWh, which at the exchange rate of 4.0 TL per euro applied throughout this paper is 0.106 EUR per kWh and gives about 15 EUR per month. Both units are stated so that the kWh and TL rows of Table A5 can be checked against each other. Because item 4 of the questionnaire covers lighting only, this figure is a lower bound for household electricity use, and the aggregate expenditure derived from it in Note 3 is correspondingly a lower bound. The monthly kWh average rests on 8 to 9 respondents per month, as reported in the note to Table A5. Since there were approximately 100,000 residences, total consumption of 5,112,000,000 kWh was obtained over a 30-year period (see Appendix A, Note 3 for calculations). This extrapolation is an explicitly conditional scenario and not a measurement of city-wide consumption: it applies the mean of eight to nine monthly answers to 100,000 households and assumes that every household would install a system. Its sensitivity is linear in the assumed consumption level. At 30% lower consumption, the thirty-year lighting bill is 378,000,000 EUR, and at 30% higher consumption, it is 702,000,000 EUR; consumption would have to be about 36% above the surveyed lighting figure, that is, about 193 kWh per month at the same tariff, for the thirty-year bill to equal the 735,000,000 EUR investment computed below. No documented willingness to purchase is claimed at any of these levels.

A 1 kWp PV system entails a set-up cost of 2450 EUR, including 250 EUR for maintenance and 200 EUR for the inverter (Appendix A, Note 1), so a 3 kWp residential system costs 7350 EUR (2450 EUR multiplied by 3). Multiplying by 100,000 residences gives a total residential PV investment for Eskisehir of 735,000,000 EUR. Note that this price refers to a complete installed system, including an inverter and maintenance, not to a single 180 Wp panel. Comparing the two magnitudes, household PV requires 195,000,000 EUR more than the thirty-year lighting electricity bill (540,000,000 EUR minus 735,000,000 EUR equals minus 195,000,000 EUR). The sign of this difference is negative, so on this basis and at these prices the investment is not privately profitable for the household. Three qualifications apply. First, both magnitudes are undiscounted, and a proper appraisal requires discounted cash flows, which are not computed here (Section 4.4). Second, the avoided-cost base is a lower bound because item 4 of the questionnaire covers lighting only; measured against total household electricity, the balance would be more favorable. Third, the comparison uses the 2013 tariff and the module prices reported in [21], and module prices have fallen

substantially since. Under these conditions, the result implies a need either for a subsidy or for a higher avoided-cost base.

Eskisehir had approximately 100,000 residences. At 21.3 m^2 of module area per 3 kWp system, the module area required by the households of the city is about 2,130,000 m^2, corresponding to 100,000 systems. Since 7350 EUR is the price of a complete 3 kWp system and not of a single panel, the market value is 100,000 multiplied by 7350 EUR, that is, 735,000,000 EUR, which by construction equals the household investment computed above. We also note that 7350 EUR is the installed-system price, including an inverter and maintenance, whereas a module manufacturer captures only the module share of that price, so the revenue magnitudes below are upper bounds. We therefore report both price variants in this section and compare them in Section 4.4, rather than discussing them only as limitations. At the assumed reference scale of 15 production workers, the plant assembles 21.2 three-kilowatt module sets per day in the ideal case and 13.0 in the benchmark case. At 21.2 sets per day, 7738 sets (21.2 × 365 days) are assembled per year in the ideal case, and at 13.0 sets per day, 4745 sets (13.0 × 365 days) are assembled per year in the benchmark case. Over the thirty-year horizon, this gives 232,140 sets (7738 × 30) in the ideal case and 142,350 sets (4745 × 30) in the benchmark case.

Assuming that the population in Eskisehir grows at a constant annual rate of 2.2% over a 30-year period, corresponding to the rate observed in 2016 [33], we would need to manufacture approximately twice the initial number of PV modules, or 193,479 PV modules (i.e., $100{,}000 \times e^{0.022 \times 30} = 193{,}479.2 \approx 193{,}479$). At 7738 sets per year, this quantity would be reached in 25 years in the ideal case. At 4745 sets per year, it would be reached in 41 years in the benchmark case, which exceeds the assumed thirty-year life of the plant. In other words, a single plant of this size covers the projected residential demand of the city within its lifetime only in the ideal case; in the benchmark case, it does not, and the shortfall would have to be met by imports, by a second plant, or by a longer horizon. Cumulative output over thirty years, 232,140 sets in the ideal case, also exceeds the projected local demand of 193,479 sets, so the surplus presupposes demand outside Eskisehir. No evidence of such demand is presented here, and the scenario should be read as conditional on it. This comparison is an internal consistency check between two assumptions, not a test against data.

The revenue obtained from 232,140 sets in the ideal case is 1,706,229,000 EUR (232,140 × 7350 EUR), and that obtained from 142,350 sets in the benchmark case is 1,046,272,500 EUR (142,350 × 7350 EUR) over the thirty-year life span. Divided by the minimum wage, these are 3,412,458 units (1,706,229,000 EUR/500 EUR) and 2,092,545 units (1,046,272,500 EUR/500 EUR), respectively. All magnitudes in this section are placed on a common thirty-year basis, and the annual figures are obtained by dividing the thirty-year totals by thirty. The labor cost is the same in both scenarios because the effectiveness index scales output without changing the assumed scale: 17 employees (15 production workers organized in three shifts and 2 support staff) at 500 EUR per month give 102,000 EUR per year and 3,060,000 EUR over thirty years, that is, 6120 wage units in each case. The monetary fixed cost is treated separately because a competitive free-market assumption and near-complete public financing of fixed capital are not compatible and must not be mixed: under the subsidy scenario the government covers 235,000 EUR per year (470 wage units) and 15,000 EUR per year (30 wage units) remains with the firm, whereas under the market scenario the firm bears the full 250,000 EUR per year (500 wage units), that is 7,500,000 EUR over the thirty-year horizon. These are monetary magnitudes and are unrelated to the scale parameter K of Equation (4). Therefore, revenue net of labor cost, which is a gross margin and not a net profit, is 3,406,338 units (3,412,458 units minus 6120 units) in the ideal case and 2,086,425 units (2,092,545 units minus 6120 units) in the benchmark case. In current prices,

this is 1,703,169,000 EUR (3,406,338 units × 500 EUR) over thirty years and 56,772,300 EUR per year in the ideal case, and 1,043,212,500 EUR (2,086,425 units × 500 EUR) over thirty years and 34,773,750 EUR per year in the benchmark case. These magnitudes are revenue minus labor cost only. They omit cells, glass, frames, encapsulants, junction boxes, and the remaining bill-of-materials items, the electricity consumed by the production line, buildings and equipment, depreciation, financing, working capital, quality control, scrap, logistics, sales, insurance, and taxes. A net profit cannot be computed without a bill of materials and a plant budget, neither of which exists for a plant that has not been built; the figures reported here are therefore gross margins over labor cost and an upper bound on profitability. For the same reason, no discounting is applied, and neither a net present value nor an internal rate of return is reported. The same calculation at the price a module manufacturer can charge is reported here rather than only in the limitations. Valuing each set at two-thirds of the installed-system price, that is, 4900 EUR, gives 1,137,486,000 EUR of revenue in the ideal case (232,140 × 4900 EUR) and 697,515,000 EUR in the benchmark case (142,350 × 4900 EUR), that is, 2,274,972 and 1,395,030 units. Subtracting the same labor cost of 6120 units from each leaves 2,268,852 and 1,388,910 units, that is, 1,134,426,000 EUR and 694,455,000 EUR over thirty years, or 37,814,200 EUR and 23,148,500 EUR per year. Both price variants are compared side by side with the other scenario variations in Section 4.4. Every figure in this section follows from the daily output rounded to 0.1 sets, which is the single rounding rule used in the paper and in the supplementary calculations. None of these magnitudes establishes the financial feasibility of a plant or the profitability of a particular investment; they are the sales and employment implications of the assumptions listed in Table 1.

The scenario results are conditional on the assumptions collected in Table 1 and on the cost items that are not included, so they are indicative rather than evidence of the profitability of an actual investment. However, there are some important considerations that governments should take into account when providing favorable investment conditions for entrepreneurs. Moreover, environmentally friendly solar energy will be used, and this would decrease Türkiye's energy (natural gas) imports. This will also increase energy independence, which is important for energy security. To assemble 21.2 module sets per day at the assumed reference scale, 15 production workers organized in three shifts and two support staff would be employed, that is, 17 persons in a single plant. We make no claim about city-wide or national unemployment: establishing wider employment effects would require a regional labor-market analysis and an input–output or multiplier assessment, which are outside the scope of this study.

4.4. Limitations and Sensitivity

The scope of what this study can support should be stated without hedging. No PV module plant operates in Eskisehir, so there are no plant-level observations of output, inputs, cycle times, scrap rates, or material flows. Every parameter in Table 1 is assumed or calibrated; none is econometrically estimated, and no standard error, residual, or test statistic can be attached to any reported magnitude. The survey characterizes potential residential demand and cannot identify labor productivity, output elasticities, or plant capacity. The results are therefore conditional scenario results, and the framework is offered as a transparent structure to be re-estimated when plant-level or panel data become available. The six scenario comparisons are summarized in Table 2.

Table 2. Scenario comparison: effectiveness, price, and elasticity variants. All six scenarios are evaluated at the same assumed reference scale: A = 1, g = 0, α = 0.2, scale parameter K = 30, reference workforce L = 15 production workers, 17 employees, 365 operating days, a thirty-year horizon, and a labor cost of 500 EUR per employee per month. The reference workforce is an assumption and no scenario, S6 included, represents a recomputed cost minimum. Daily output is rounded to 0.1 sets and every monetary magnitude follows from the rounded value, on a common thirty-year basis. S1 to S6 are the labels of the six scenarios compared in the table, and each is defined in the first column. The gross margin is revenue minus labor cost only, as defined in Section 4.3.2.

Scenario	H	Workers	Sets Per Day	Price Per Set (EUR)	Gross Margin Per Year (EUR)	What Changes; Everything Else As Stated
S1 Ideal case, installed-system price	1.000	15	21.2	7350	56,772,300	Reference scenario of Section 4.3.2
S2 Benchmark case, installed-system price	0.611	15	13.0	7350	34,773,750	H only; scale, energy, capital, price and horizon as in S1
S3 Ideal case, producer price	1.000	15	21.2	4900	37,814,200	Price only; quantities, labor and horizon as in S1
S4 Benchmark case, producer price	0.611	15	13.0	4900	23,148,500	Price only; quantities, labor and horizon as in S2
S5 Effectiveness raised by 10 percentage points	0.711	15	15.1	7350	40,407,525	H only; output rises by 16.4%, not by 10%
S6 Output elasticities β = 0.4, γ = 0.4	0.611	15	12.1	7350	32,359,275	Elasticities only, evaluated at the same scale L = 15; α = 0.2

Sensitivity to the main assumptions runs as follows. Output is proportional to the effectiveness index H, so the effect of a change in H is proportional to, and not equal to, the change in H: raising H from 0.611 to 0.711, that is, by 10 percentage points, raises output and revenue by $0.711/0.611 - 1 = 16.4\%$ when every other input is unchanged (Table 2, scenario S5). Percentages and percentage points are distinguished throughout the paper. Varying the output elasticities by 0.1 within the constant-returns restriction changes the output obtained at the reference workforce but leaves the qualitative shape of the production and cost curves unchanged; scenario S6 of Table 2 reports this variant at the same reference scale L = 15. If output is valued at the module share of the installed-system price rather than at the full system price, that is, at two-thirds of it, the annual gross margin falls from 56,772,300 to 37,814,200 EUR in the ideal case and from 34,773,750 to 23,148,500 EUR in the benchmark case (Table 2, scenarios S3 and S4). If residential systems are sized on daily energy rather than on peak power, that is, at about 0.56 kWp on the illustrative six-hour assumption or about 0.83 kWp on the irradiation-based one instead of 3 kWp, the investment per household, the market volume, and the revenue figures all fall by a factor of about five or about four respectively (Section 4.3.1). These four dimensions, rather than the arithmetic of any single scenario, determine the order of magnitude of the results.

The data have further limitations. The survey dates from 2013 and is retained because it is the only firsthand household dataset available to the authors for Eskisehir; it is used as a historical demand illustration rather than as a current market measurement, and repeating it with a documented probability sample is the first priority for future work. The sample is a convenience sample, the number of valid answers is small for the items reported in kWh, and the questionnaire contains no income or occupation item, so no claim of socio-economic representativeness is made. Item 4 covers lighting only, which makes the avoided-cost base a lower bound. The irradiation assumption rests on the official atlas value for Eskisehir together with a single-day spot measurement from 1986, which cannot represent thirty years of production; a long-term meteorological series with tilt, orientation,

temperature, and loss corrections would be required for a production estimate. Finally, the model combines sources from 1986, 1998, 2013, 2015, and 2025, and monetary values are not deflated to a common base year or adjusted for exchange-rate movements.

On the financial side, the reported magnitudes are revenue minus labor cost. A bill of materials, a plant budget, depreciation, financing, working capital, quality control, scrap, logistics, insurance, and taxes are not included, so no net profit, net present value, or internal rate of return is computed, and no discount rate is applied. The scenarios assume 365 operating days per year without maintenance shutdowns or holidays, and they do not model the continuing decline in module prices, which would reduce both the investment cost per household and the revenue per set.

The study quantifies economic magnitudes only. Life-cycle assessment, carbon footprint, energy payback time, water use, waste and recyclability, critical materials, supply-chain local content, and occupational safety are not measured here. The environmental statements in this paper are therefore qualitative, and the title and the conclusions have been narrowed accordingly; quantified sustainability indicators are identified as a separate study.

5. Conclusions

This study develops economic scenarios of production, employment, and sales for solar PV module manufacturing based on local resources, which may contribute to technological improvement for developing countries. In this regard, the widespread use of PV module manufacturing can contribute to the development of an effective production process in Türkiye. To this end, we specified and calibrated a production function to characterize the manufacturing process of a hypothetical solar PV module plant over a thirty-year horizon in Eskisehir. The parameters are calibrated rather than estimated econometrically. We included the energy consumption inputs of the firm in a Cobb–Douglas production function, with the necessary subsidization provided by the government. We show a potentially realistic situation and an ideal case for a PV module manufacturing firm in Eskisehir using our approach, with assumed efficiency levels of 61.1% and 100%, respectively. In the benchmark case of 61.1% effectiveness, revenue net of labor cost is 34,773,750 EUR per year at the assumed reference scale; in the ideal case of 100% effectiveness, it is 56,772,300 EUR per year. Valued at the module share of the installed-system price, the same two scenarios yield 37,814,200 EUR and 23,148,500 EUR per year (Table 2). These are gross margins over labor cost and not net profits: material, energy, capital, financing, and logistics costs are not included, and the magnitudes are undiscounted. None of them demonstrates the financial feasibility of a plant or the profitability of a particular investment. The contribution of the paper is therefore a transparent and reproducible scenario framework rather than an estimated description of an existing production process.

Meanwhile, this paper theoretically shows that there is a relationship between "Leisure time" and "Work" by expressing these terms in relation to the determination of "Output". This is achieved by drawing an analogy between the output term of the production function and the physical concept of work. The construction records an assumed or measured effectiveness level in a compact normalized form; it does not measure that level, which must come from firm-level data. Its usefulness is therefore expository and organizational rather than evidential. Any such measurement would have to come from firm-level data on the production factors; the framework organizes the comparison but does not supply the measurement. It is shown that when human capital skills and R&D activity levels decrease in high-technology manufacturing firms under the conditions of Eskisehir, this situation negatively affects the production level of a firm through the channel of efficiency

and productivity and establishes a mathematical relationship between "Leisure time" and "Work", which can be predicted by our proposed equation.

The cos θ term records an assumed effectiveness level in a compact form. It does not by itself make production decisions more accurate; what the construction offers is a consistent way of comparing scenarios at different assumed effectiveness levels. By construction, zero effectiveness corresponds to $\theta = 90^{\circ}$, where no effective work is performed, and maximum effectiveness corresponds to $\theta = 0^{\circ}$, consistent with Table A3. Setting out these indicators for a PV module firm makes the assumptions behind a production decision explicit; it does not by itself establish what such a decision should be. Subsidy design and employment policy would require the regional labor-market and multiplier analysis identified in Section 4.4, which this study does not provide.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/su182110859/s1. Supplementary File S1 (Survey-Data.xlsx): An Excel workbook containing the original household survey records in Turkish and English worksheets, which were used to calculate the values reported in Table A5. Supplementary File S2 (Calculations.xlsx): An Excel workbook containing the model parameters and production, cost, reference-point, conversion, and scenario calculations underlying Figures 6 and 7. All scenario results reported in Section 4.3.2.

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Institutional Review Board Statement: The household survey was conducted in 2013 at Eskisehir Osmangazi University. Participation was voluntary and anonymous; the questionnaire collected no personal identifiers, sensitive personal data, or health-related information, and the study was not clinical or interventional. It therefore meets the requirements for ethical exemption set out in the Scientific Research and Publication Ethics Directive of Eskisehir Osmangazi University (available at https://ogu.edu.tr/Icerik/Index/309/bilimsel-arastirma-ve-yayin-etigi-kurulu (accessed on 16 October 2026), with its full text published in the Turkish Higher Education Council database at https://kms.kaysis.gov.tr/Home/Goster/172919 (accessed on 16 October 2026)) for social science studies that collect anonymous, voluntary survey data, and thus did not require ethical approval. The documentation of this institutional framework was submitted to the editorial office with the first submission.

Informed Consent Statement: Oral informed consent was obtained from every respondent before the questionnaire was completed. Respondents were informed of the purpose of the survey, of the anonymity of their answers, and of their right to decline. No personal identifiers were recorded, and the consent script used during data collection is available from the authors.

Data Availability Statement: The raw survey records are provided as Supplementary Materials in Survey-Data.xlsx, in Turkish and English sheets, and all entries of Table A5 are computed from that file. The calculations underlying Figures 6 and 7 and all scenario results reported in Section 4.3.2 are available from the authors on request.

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Appendix A

Table A1. The Relationship between the share of innovation capacity and R&D personnel in Ankara, Türkiye.

		The Share of R&D Personnel			
		0 (None)	**0–0.1 (Low)**	**0.1 + (High)**	**Total**
No innovation capacity	number	23	6	8	37
	horizontal percentage *	62.2 #	16.2	21.6	100.0
	vertical percentage **	53.5 ##	35.3	22.2	38.5
Low innovation capacity	number	6.0	4.0	6.0	16.0
	horizontal percentage	37.5	25.0	37.5	100.0
	vertical percentage	14.0	23.5	16.7	16.7
High innovation capacity	number	14	7	22	43
	horizontal percentage	32.6	16.3	51.2	100.0
	vertical percentage	32.6	41.2	61.1	44.8
Total					
	number	43	17	36	96
	horizontal percentage	44.8	17.7	37.5	100.0
	vertical percentage	100.0	100.0	100.0	100.0

Source: [22]. **Note:** * Horizontal percentage: Example for the #: (23/37) × 100 = 62.2, ** vertical percentage: Example for the ##: (23/43) × 100 = 53.5.

Table A2. Substitution elasticities reported by [23] for Germany. These values are reproduced as a qualitative reference only and are not used as parameter values in this study.

	σα1	**σβ1**
(K, E)/L	0.653 [(K, E)/L]	0.846 [(K, E)/L]
(K, L)/E	0.822 [(K, L)/E]	0.458 [(K, L)/E]
(E, L)/K	0.422 [(E, L)/K]	0.146 [(E, L)/K]

Source: [23].

Table A3. Various productivity combinations by θ angle.

θ	**cos θ**	**F × d × cos θ (E × L × K × cos θ)**
0°	1	1.000Fd (max productivity)
30°	0.866	0.866Fd
45°	0.707	0.707Fd
60°	0.500	0.500Fd
90°	0	0.000Fd (zero effectiveness)

Table A4. The survey questions about electricity consumption.

1. What is the number of your family members in your household?
2. What type of house do you stay in?
2.1. Apartment flat
2.1.1. If you stay in an apartment flat, how many flats are there in your apartment?
2.2. Detached house

Table A4. *Cont.*

3. What type of equipment that use electricity do you have in your residence, and how many of each item? 3.1. Television 3.2. Washing machine 3.3. Refrigerator 3.4. Vacuum cleaner 3.5. Computer 3.6. Toaster 3.7. Any others. . .
4. How much energy do you consume in your residence for lighting purposes per month, in kWh or TL?
5. What type of heating system do you use in your residence for warming purposes? 5.1. Combi boiler 5.2. Central heating system 5.3. Electrical heating system 5.4. Diesel heating system 5.5. Others
6. Do you use air conditioning for cooling in your residence? 6.1. If you have a "yes" answer to question 6, how much energy do you consume in your residence in kWh or TL for cooling purposes of your residence for each month?

Source: From the author's firsthand collected survey, 2013.

Table A5. Statistics on the responses to the survey questions.

Questions/ Units	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.	Av.
(4) kWh	160.75	151.25	148.25	139	145.38	130	129.38	128.25	136.62	140	153.12	146.78	142.40
(4) TL	64.20	64.28	61.61	59.44	58.25	57.55	56.96	56.53	59.59	59.84	63.38	65.37	60.58
(6.1) kWh	-	-	-	-	-	45.60	102.60	123.51	27.36	34.20	-	-	27.77
(6.1) TL	-	-	-	-	-	20	45	54.17	12	15	-	-	12.18

Source: From the author's firsthand collected survey, 2013. TL denotes Turkish lira. Note: Entries are means over the respondents who answered the item, computed from the final dataset (Survey-Data.xlsx, English sheet, rows 2 to 146, n = 145 records). The number of valid answers for item 4 is 8 per month in kWh and 9 in December, and 142 per month in TL and 141 in December; for item 6.1 in TL it is 3 in June, 6 in July, 6 in August, 5 in September and 1 in October. The dataset contains no direct kWh answers for item 6.1, so the kWh entries for that item are derived from the TL answers with a single conversion constant of 2.28 kWh per TL, which corresponds to the 2013 residential tariff of about 0.44 TL per kWh and reproduces the relation between the two units observed for the months with answers; for example, 12.00 TL in September gives 27.36 kWh. The annual mean in the last column is the sum of the twelve monthly values divided by twelve, with months in which no cooling was reported entered as zero. Entering zero for those months is a computational assumption and not an observation: the questionnaire returned no answers for them, and we do not know whether the households used no cooling or simply did not report it. A dash in the table denotes such a month with no answer, which is different from a reported value of zero; no respondent reported zero cooling consumption. On this assumption the annual means are 27.77 kWh and 12.18 TL for item 6.1, and they are lower bounds if any of the unanswered months carried positive consumption.

Note 1:

Some calculations and a possible scenario:

In total, 2190 kWh per year of energy is obtained (i.e., 1 kWp × 6 h/day × 365 days = 2190 kWh/year). However, there will be some cable and inverter energy losses. These energy losses are assumed to be 10% on average. Therefore, 1971 kWh per year (i.e., 2190 kWh × 0.9 = 1971 kWh/year) remains after losses. The monthly and daily total consumption amounts are shown on your electricity bill. If a household consumes 10 kWh on average per day, annual consumption is 3650 kWh (i.e., from 10 kWh/day × 365 days = 3650 kWh per year). In this case, a 1 kWp PV system covers about 54% of annual household

electricity consumption [21]. Electricity was priced at 0.32 TL per kWh in 2013; the tariff then applied to commercial users such as hotels. At the exchange rate of 4.0 TL per euro applied throughout this paper, that tariff is 0.32/4.0 = 0.08 EUR per kWh. The avoided cost has to be computed on the energy actually delivered, that is after losses, so the annual saving is 1971 kWh per year × 0.08 EUR per kWh = 157.7 EUR per year. For a 1 kWp PV system, the materials cost 2000 EUR, maintenance 250 EUR, and the inverter 200 EUR, so 450 EUR is required for set-up and regulation on average. The total cost is now 2450 € (i.e., 2000 € + 450 € = 2450 €), so on this assumption the payback time is 2450/157.7 = 15.5, about 16 years. The same calculation on the irradiation-based assumption of 4.03 equivalent full-power hours per day gives, for a 1 kWp system, an annual generation numerically equal to the atlas value itself, 1472 kWh per year before losses and 1472 × 0.9 = 1325 kWh after 10% losses. At the same tariff and exchange rate the annual saving is 1325 × 0.08 = 106.0 EUR per year, and the payback time is 2450/106.0 = 23.1, about 23 years. The 16-year figure is therefore the illustrative upper case and the 23-year figure the irradiation-based one. Both are computed on the energy delivered after losses, and neither includes degradation, discounting or exchange-rate movements. In addition, hotels that produce their own electricity may benefit from renewable energy laws in the near future. For instance, in the European Union, Holland and Germany currently gain $ 0.33 for 1 kWh from the government because they produce their own electricity. In line with these regulations and 2023 targets in the country, Türkiye will also benefit from these regulations [21].

Note 2:

A residential system sized on peak power at 3 kWp requires about 21.3 m^2 of module area (3 kWp × 7.1 m^2 per kWp); a 6 kWp system would require about 42.6 m^2. At 2450 EUR per kWp, the 3 kWp system costs 7350 EUR (2450 EUR × 3 kWp), and the module is warranted to deliver at least 80% of its nameplate power after 20 years [21]. This amount of money will be given only once, and no further payment will be made after that [21]. For comparison, a system sized on daily energy rather than on peak power would require 3 kWh per day/(6 h per day × 0.9) = 0.56 kWp on the illustrative assumption, or 3 kWh per day/(4.03 h per day × 0.9) = 0.83 kWp on the irradiation-based one, that is about 4 m^2 and about 6 m^2 of module area and a set-up cost of about 1370 EUR and about 2030 EUR at 2450 EUR per kWp.

Note 3:

A total of 5,112,000,000 kWh (i.e., 142 kWh × 100,000 × 12 × 30 = 5,112,000,000 kWh) or 540,000,000 € (i.e., 15 € × 100,000 × 12 × 30 = 540,000,000 €) is obtained, the monthly figure of 15 € being 142 kWh at the residential tariff of 0.106 EUR per kWh derived in Section 4.3.2.

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